

Pranav Kandpal

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EDUCATION

Dronacharya College of Engineering — Gurugram University <i>Bachelor of Technology in Computer Science and Engineering — CGPA: 8.2 / 10</i>	Gurgaon, India 2023 – 2027
Kamal Model School — CBSE <i>Grade 12: 93% Grade 10: 77%</i>	Delhi, India 2021 – 2023

EXPERIENCE

Doctor's EHR LLC <i>Software Development Intern Java, Spring Boot, MinIO, JWT</i>	Remote June 2025 – July 2025
- Built RESTful API endpoints in Java Spring Boot for the Patient Management module, integrating MinIO (S3-compatible storage) for secure, scalable patient document uploads. - Designed a centralized error-handling framework with custom exception pipelines, standardizing API responses and improving backend reliability across all service modules. - Implemented JWT-based authentication with server-side input validation to harden patient data access and reject malformed requests at the API boundary. - Authored a Postman test suite covering 6+ core modules, reducing frontend integration friction by providing a definitive API contract.	

PROJECTS

RetailPulse (AI Forecasting System) Python, XGBoost, RAG, Streamlit - Built a retail forecasting platform combining XGBoost with RAG pipelines to generate interpretable forecasts under SOP constraints. - Engineered feature pipelines using lag variables, rolling statistics, and seasonal transformations to improve forecasting accuracy and reduce temporal leakage. - Implemented a tiered interpretation system using Llama-3 (Groq) for visual explanation, risk analysis, and operational recommendations.
FluxDB C++17, TCP/IP Networking, Concurrency - Architected a multi-threaded in-memory NoSQL database with a custom binary TCP/IP protocol for low-latency persistent client-server communication. - Designed an adaptive indexing engine switching between Hash and Sorted Indexes based on query patterns, improving mixed-workload retrieval efficiency. - Implemented WAL-based crash recovery and published a native Python driver on PyPI (fluxdb-driver) for lightweight TCP communication.
Sift (Data Quality Engine) Python, Scikit-Learn, Polars - Built a deterministic data quality engine to detect schema drift, anomalous distributions, and inconsistent labelled data. - Implemented Isolation Forest anomaly detection and N-Gram clustering for fuzzy duplicate resolution across structured datasets. - Designed a chaos-testing framework injecting synthetic corruption to benchmark pipeline robustness and anomaly recall performance.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, Java (Spring Boot), SQL
Backend & Systems: TCP/IP Sockets, Client-Server Architecture, Async I/O, Multi-threading, Memory Management, REST APIs
Data & AI: XGBoost, Scikit-Learn, RAG Architectures, LangChain, ChromaDB, Time-Series Forecasting
Infrastructure & Tools: Docker, Git, Linux (Bash), CMake, GDB, Valgrind, MinIO, PyPI Publishing
Systems Concepts: WAL, ACID Persistence, Binary Protocol Design, Distributed Systems